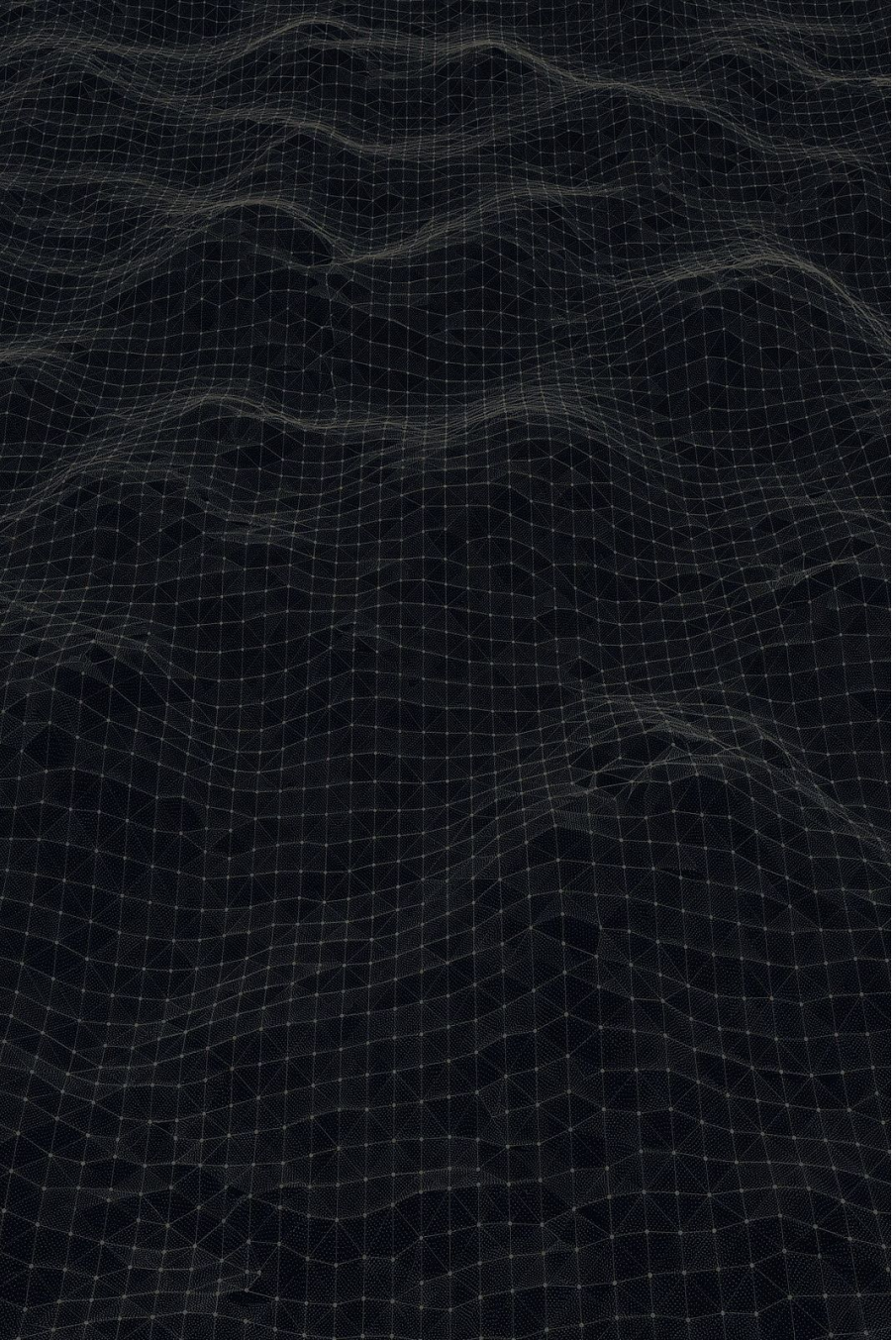
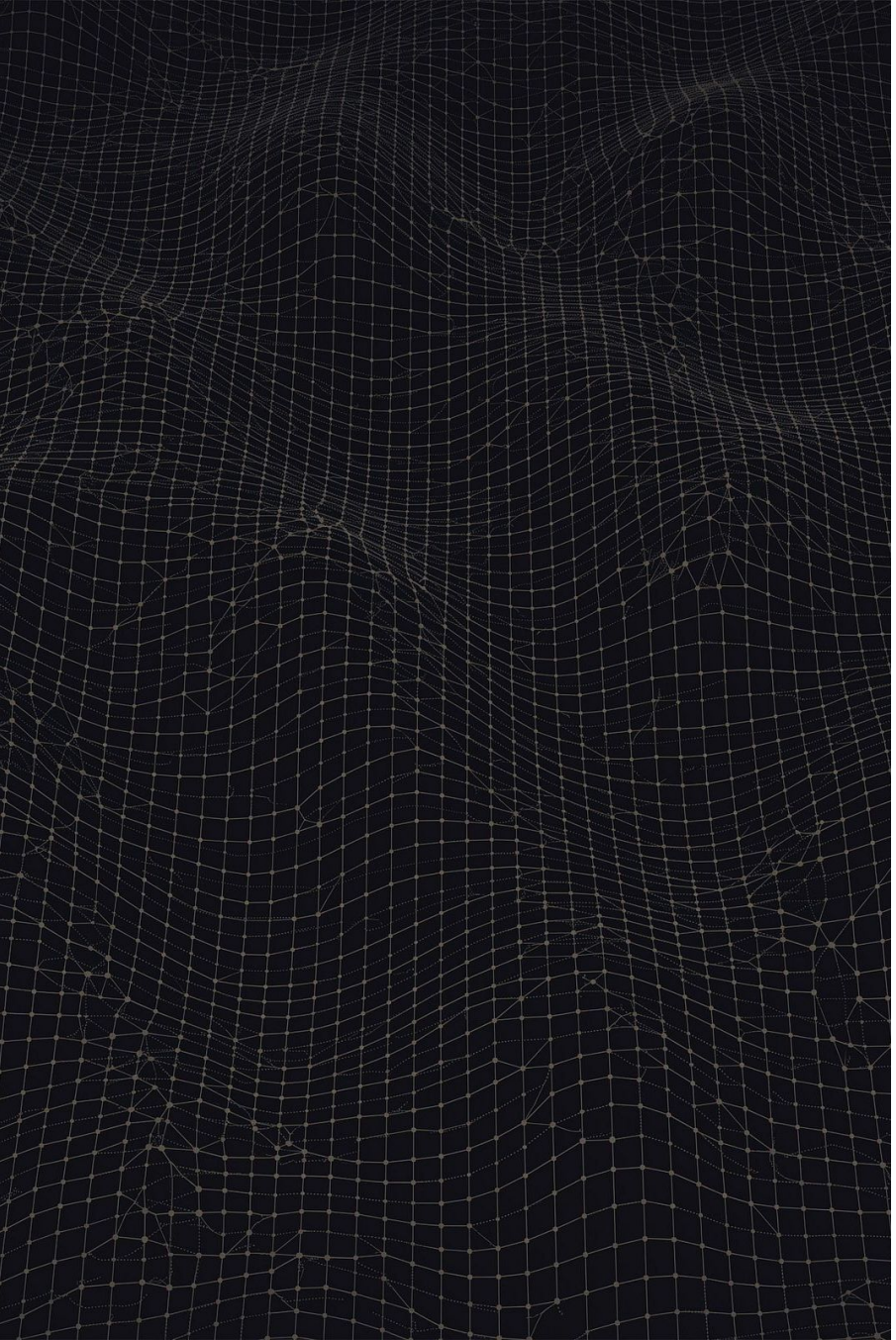


Bias and Fairness in Machine Learning

Identifying and Mitigating Biases in ML Models

A good ML model is not only accurate — it should also be reliable, explainable, responsible, and fair.

Dr. Anilkumar V Brahmane
Computer Engineering Department | Sanjivani College of Engineering



What is Bias in ML?

Simple Definition

Bias means a model systematically gives better or worse results to some people or groups because of patterns in the data, model, or decision process.

Classic Example: Job Recruitment

A company trains an ML model on historical hiring data where mostly men were hired. The model learns: **"Male candidates are more likely to succeed."**

Two equally qualified candidates:

- Candidate A → Male → **Selected**
- Candidate B → Female → **Rejected**

This is **bias** — not based on merit, but on historical patterns.

Dr. Anilkumar V Brahmane | Computer Engineering Department | Sanjivani College of Engineering

Bias vs. Fairness — Core Concepts

⚠️ Bias

Unwanted systematic preference or disadvantage toward a group.

✅ Fairness

Treating groups appropriately and without unjustified discrimination.

🚫 Discrimination

Unfair treatment based on gender, race, age, or other characteristics.

🎯 Fair ML

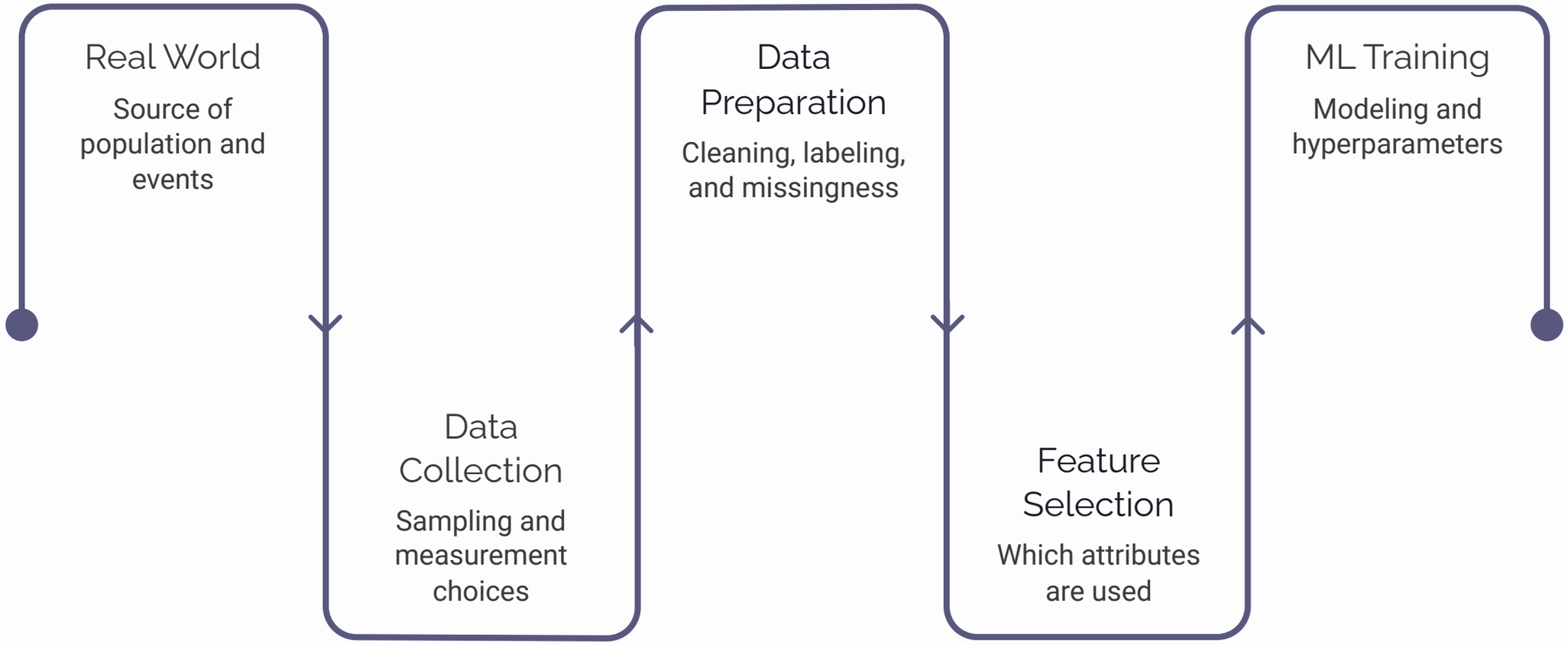
Designing ML systems to reduce unfair outcomes systematically.

Key Principle: Bias is the problem. Fairness is the goal. The aim is not identical predictions for everyone — it is to prevent unjustified systematic disadvantage.

Dr. Anilkumar V Brahmane | Computer Engineering Department | Sanjivani College of Engineering

Where Does Bias Enter the ML Pipeline?

Bias can enter **at any stage** of the ML workflow — not just in the algorithm itself.



Understanding this pipeline is critical: fixing only the algorithm is not enough. We must audit every stage — from data collection to the final human decision.

Types of Bias — Part 1

1. Historical Bias

Comes from existing inequalities in society. Example: If a company historically hired 90 men and 10 women, the ML model learns "male = successful." The problem is in the data, not just the algorithm.

2. Sampling Bias

Training data does not represent the real population. Example: A face recognition system trained on 8,500 Group A images vs. 500 Group C images achieves 99% accuracy for Group A but only 75% for Group C — yet reports 96% overall.

3. Measurement Bias

Data does not measure the concept correctly for all groups. Example: Using *healthcare spending* as a proxy for *healthcare need* disadvantages patients with low income or poor hospital access.

4. Label Bias

Output labels themselves contain bias. Example: "Student success" labeled by subjective teacher ratings. Biased human judgment → biased labels → biased model predictions.

Types of Bias — Part 2

5. Algorithmic Bias

The model optimizes overall accuracy without considering group-level fairness. Group A = 98% accuracy, Group B = 80% accuracy — the model looks excellent overall but is unfair.

6. Representation Bias

Some groups are poorly represented. A voice assistant trained 70% on Accent A and 10% on Accent C will struggle with Accent C users. Affects speech recognition, transcription, and customer-service bots.

7. Proxy Bias

A sensitive attribute (e.g., race) is not used directly, but another feature (e.g., ZIP code) indirectly reveals it. Removing gender from a loan model does NOT remove bias if ZIP code acts as a proxy.

8. Feedback Loop Bias

Predictive policing example: Model flags Area A as high-risk → more police sent → more incidents recorded → model flags Area A as even riskier. The prediction reinforces itself in a dangerous loop.

Bias is not limited to gender or race — it can involve environments, languages, accents, lighting conditions, age groups, and more.

How to Identify Bias in Your Model

Step 1: Check the Dataset

Are all important groups represented?

Group	Training Samples
Group A	8,000
Group B	1,500
Group C	500 ⚠️

Only 500 samples for Group C should immediately raise a red flag.

Step 2: Compare Group Performance

Never rely on overall accuracy alone.

Metric	Value
Overall Accuracy	94%
Group A	98%
Group B	95%
Group C	78% ⚠️

Step 3: Use Confusion Matrix Per Group

For each group, calculate:

- **True Positive (TP)** — correctly approved
- **False Positive (FP)** — wrongly approved
- **True Negative (TN)** — correctly rejected
- **False Negative (FN)** — wrongly rejected

Then compare TP, FP, FN, TN values **across groups** to find where the model fails unfairly.

> **Key Rule:** Always evaluate model performance separately for each important group.

Fairness Metrics Explained Simply



Demographic Parity

Different groups should have similar positive prediction rates.
Group A: 60% approved | Group B: 58% approved ✅
Rates are relatively close — this is good.



Equal Opportunity

Compare True Positive Rate (TPR) across groups. Qualified candidates should be identified equally.
Group A: 90% | Group B: 89% ✅ vs. Group A: 95% | Group B: 60% ❌



Equalized Odds

Considers both TPR and FPR across groups.
Group A: TPR 90%, FPR 8%
Group B: TPR 88%, FPR 9% ✅ — relatively similar performance.



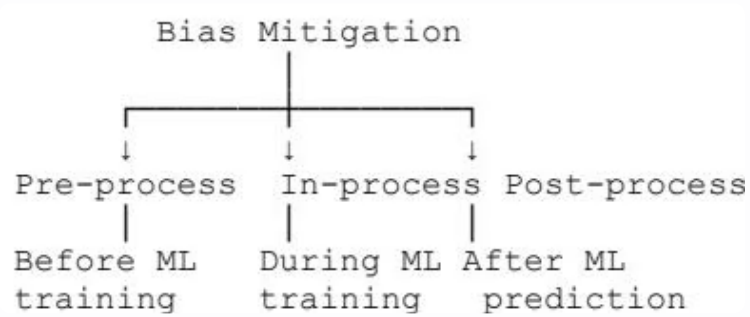
Individual Fairness

Similar individuals should receive similar predictions. If two people differ only in a sensitive attribute (e.g., gender) with identical qualifications, their predictions should not drastically differ.

Important: There is no single fairness definition that works for every application. Fairness must be defined according to the application, stakeholders, risks, and legal/ethical requirements.

Dr. Anilkumar V Brahmane | Computer Engineering Department | Sanjivani College of Engineering

How to Mitigate Bias — Three Stages



1

Pre-processing

Before ML training. Fix the data: improve data collection, re-sampling (oversample minority groups), re-weighting (give underrepresented samples higher weight), remove or modify biased/proxy features.

2

In-processing

During ML training. Modify the learning process itself. Optimize for *Accuracy + Fairness* together as a combined objective. Advanced: Adversarial Debiasing — a second model tries to detect sensitive info, forcing the main model to hide it.

3

Post-processing

After ML prediction. Adjust decision thresholds or predictions after training. Perform fairness analysis on outputs and tune thresholds per group to achieve fairer outcomes without retraining.

Pre-processing Techniques in Detail

Improve Data Collection

Collect more representative data across all groups.

Before: Group A → 90%, Group B → 10%

After: Group A → 55%, Group B → 45%

Re-sampling

Oversampling: Increase minority group samples (B: 1,000 → 4,000)

Undersampling: Reduce majority group samples (A: 8,000 → 4,000)

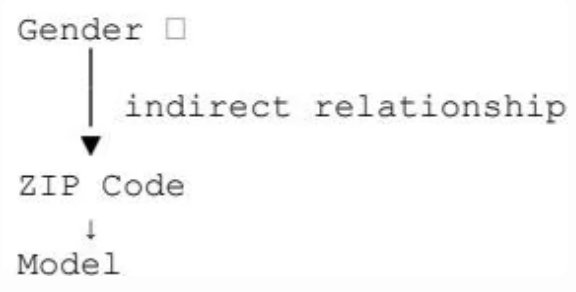
Synthetic data: Generate additional representative examples.

Re-weighting

Group A sample → weight 1.0
Group B sample → weight 2.0
Model pays more attention to underrepresented examples.

⚠️ Warning: Removing Sensitive Attributes Is Not Enough

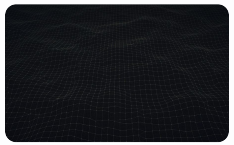
Simply removing "Gender" from a model does NOT always remove bias — because of **proxy features**.



Gender → (indirect relationship) → ZIP Code → Model

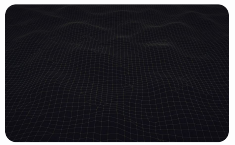
ZIP code may be strongly correlated with demographic characteristics. You must examine all correlations and proxy variables carefully.

Industry Scenarios: Bias in the Real World



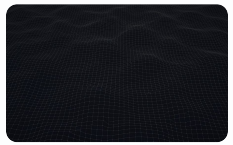
AI Recruitment

50,000 applications processed by ML. Group A shortlisted 70% vs. Group B only 35%. Root cause: historical training data showed Group A hired 80% of the time. Mitigation: audit data, remove proxy features, re-weight, fairness testing, human review.



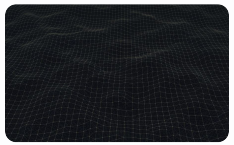
Loan Approval

ZIP code used as a feature indirectly encodes socioeconomic/demographic info. Fairness audit must compare approval rates, false rejection rates, TPR, and FPR across relevant groups.



Healthcare AI

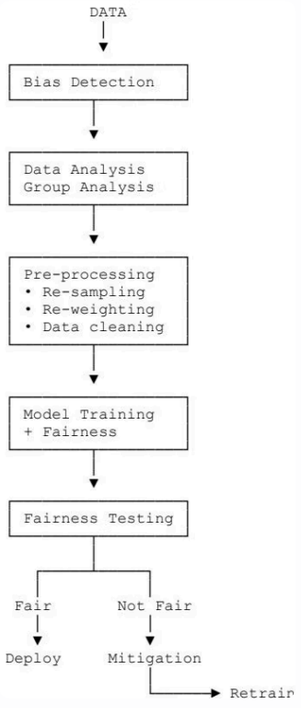
Using historical healthcare spending as a proxy for medical need disadvantages patients with low access. Fix: use clinical variables — symptoms, lab results, diagnosis, vital signs — instead of spending as proxy.



Facial Recognition

Group A: 99%, Group B: 96%, Group C: 82% — overall reported as 96%. The aggregate hides Group C's poor performance. Proper fairness evaluation must report accuracy per demographic group.

The Complete Bias Detection & Mitigation Pipeline



Pipeline Stages

01

Bias Detection

Analyze data and group representation.

02

Pre-processing

Re-sampling, re-weighting, data cleaning.

03

Model Training + Fairness

Optimize accuracy AND fairness together.

04

Fairness Testing

If Fair → Deploy. If Not Fair → Mitigate → Retrain.

05

Continuous Monitoring

Audit fairness regularly after deployment.

Critical Concepts: Accuracy ≠ Fairness

🔑 Concept 1: High Accuracy ≠ High Fairness

Model Accuracy = 95%

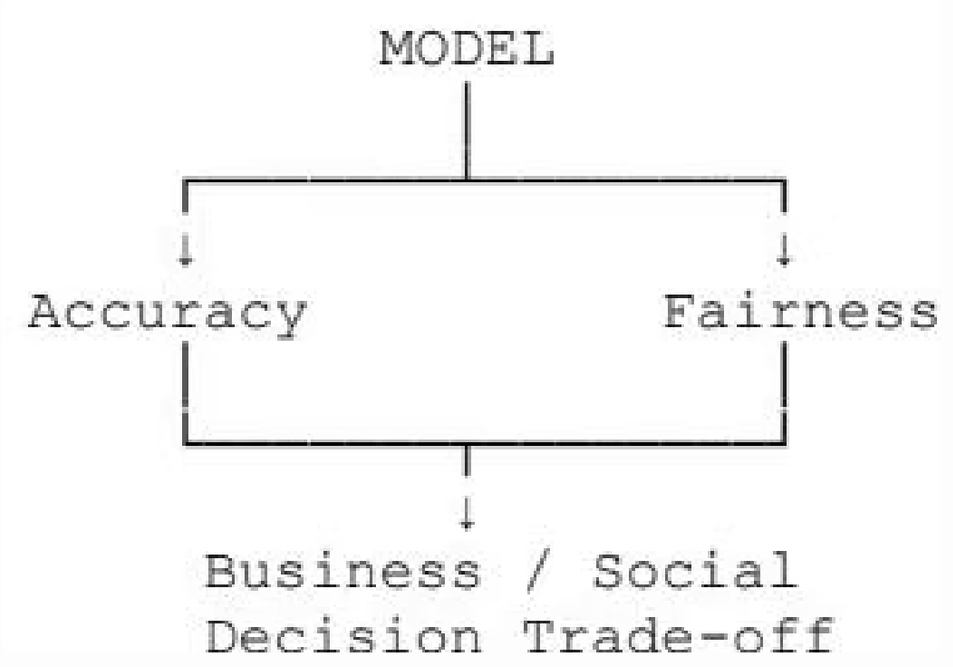
- Group A = 99%
- Group B = 78% ⚠️

Overall accuracy **hides group-level problems**. Always evaluate per group.

🔑 Concept 2: Fairness ≠ Same Outcome

Equal approval rates (50%/50%) do not automatically mean fairness. Ask: *Why are outcomes different?* Is it due to legitimate factors, poor data, historical discrimination, proxy variables, or model errors? Fairness requires **context and justification**.

Fairness–Accuracy Trade-off



ML engineers must balance multiple objectives:

- Accuracy & Fairness
- Safety & Privacy
- Cost & Interpretability
- Legal & Ethical requirements

Human-in-the-Loop

For high-impact decisions (recruitment, healthcare, loans, legal systems), ML should produce a risk score — but **high-risk cases must go through human review** before a final decision is made.

Classroom Activity & Simple Analogy

🎓 10-Minute Classroom Activity

An AI selects students for an internship. 100 students apply (50 Group A, 50 Group B).

Model selects: Group A → 30 (60%) | Group B → 15 (30%)

Q1: Is the model automatically biased? *Not necessarily — investigate first.*

Q2: What additional info do we need?

- Training data, features, labels
- Confusion matrix, TPR, FPR
- Sensitive/proxy attributes

This creates excellent classroom discussion about what fairness really means.

💡 Simple Real-Life Analogy

A teacher trains an AI using historical competition selections:

- Boys selected: 80 | Girls selected: 20

The AI learns: *"Boy → likely selected. Girl → less likely."*

Ask students: **"Is the AI intelligent?"** → Yes.

Then ask: **"But is the AI fair?"** → Now they understand the difference.

Key Lesson: AI learns patterns from data. If the data contains unfair patterns, AI may learn and reproduce those patterns.

The D→M→E→F→R Formula

D — Data

Check the data for representation and quality.

M — Model

Check how the model is trained.

E — Evaluate

Measure performance by group.

F — Fairness

Check fairness metrics explicitly.

R — Reduce

Apply mitigation techniques.

Final Takeaways for Students

1 ML learns from data

Data can contain bias — and a model can learn and reproduce that bias.

2 Overall accuracy can hide group-level problems

Fairness must be measured explicitly, not assumed from aggregate metrics.

3 Bias can be mitigated at every stage

Pre-processing, in-processing, and post-processing all offer tools to reduce unfairness.

4 AI systems must be continuously monitored

The real world changes — fairness audits must continue after deployment.

> **"Don't ask only: Is my ML model accurate? Also ask: For whom is it accurate, and for whom does it fail?"**

Dr. Anilkumar V Brahmane

Computer Engineering Department | Sanjivani College of Engineering

